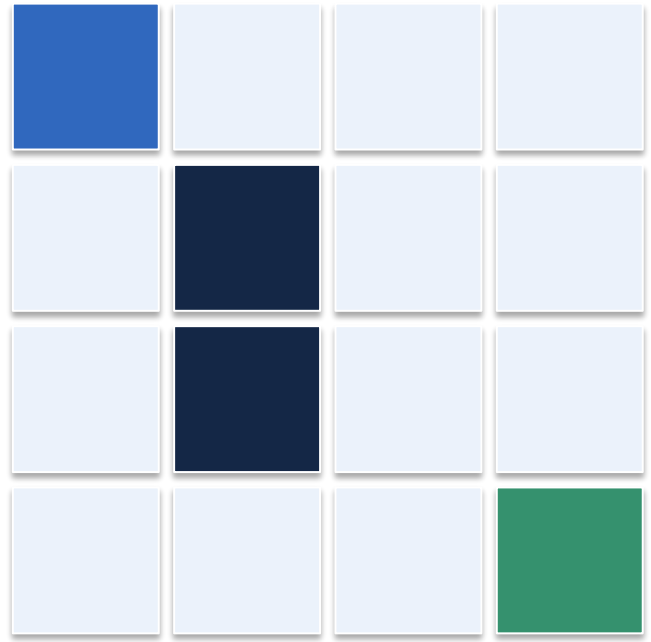
Robot Navigation with Q-Learning

Can an agent learn to reach a goal in a grid maze?

This project tests whether reinforcement learning can produce a reliable, short route in a simple environment.

Project 14.

The agent learns by trying moves and receiving rewards or penalties.



Blue: start Green: goal Dark: blocked

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Problem and Environment

MAZE

A 4 × 4 grid. The agent starts at (0, 0), aims for (3, 3), and cannot pass through two blocked cells.

STATE

The current cell tells the agent where it is.

ACTIONS

Move up, down, left, or right.

REWARD DESIGN

+10 for reaching the goal: encourages task completion.

−1 per move: encourages a shorter route.

−5 for hitting a wall or obstacle: discourages invalid moves.

An episode ends at the goal or after 50 moves.

How Q-Learning Learns

Each state-action pair gets a Q-value: an estimate of how useful an action is from that cell.

After a move, the agent updates that estimate using:

new Q = old Q + α × (reward + γ × best future Q − old Q)

Here α = 0.2 controls update size and γ = 0.95 values future rewards.

TRAINING

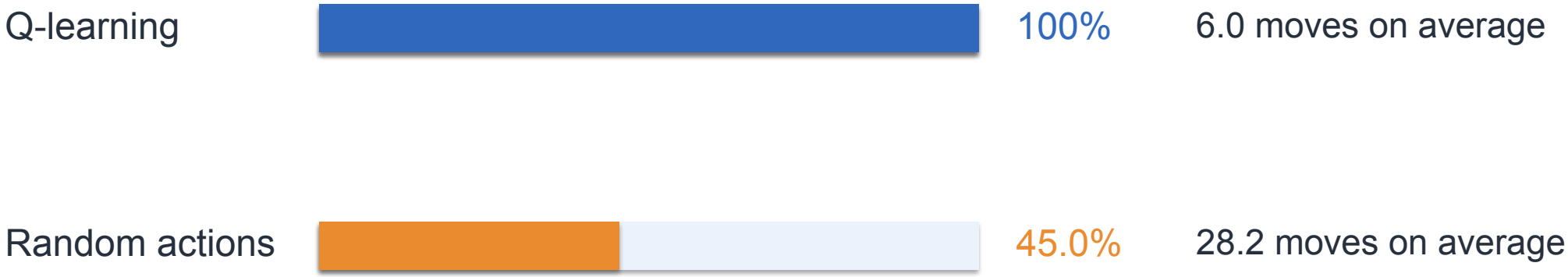
Across 1,000 episodes, ε-greedy exploration (ε = 0.2) sometimes tries a random move and otherwise uses the best-known move.

COMPARISON

We evaluate the learned policy and a random-action baseline for 200 episodes each, with the same maze and 50-move limit.

Results and What They Mean

Evaluation results: 200 episodes per agent



Interpretation: Q-learning's mean return improved early and stayed well above the random baseline. In the final evaluation, it reached the goal in all 200 episodes, averaging 6 moves. The random agent succeeded in 90 of 200 episodes (45.0%), averaging 28.2 moves when successful.

Limitations, Ethics, and Conclusion

LIMITATIONS

- Small, fixed maze; results may not generalize to larger or changing environments.
- Random-agent results vary across runs.
- Checkpoint evaluations began after the agent had already learned a route.

ETHICS AND SAFE USE

- A toy maze has low stakes; real robots can damage property or injure people.
- Test in simulation first, add safety constraints, and keep human oversight.

CONCLUSION

Q-learning performed better than random movement in this experiment.